

How to Collect Windows Defender and system Logs in Wazuh SIEM

A- Collecting Windows Defender Logs

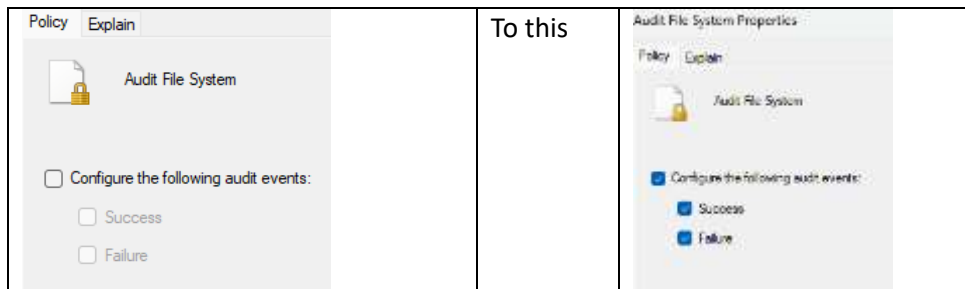
Collecting Windows Defender logs is important because it provides a critical data source for malware detection, endpoint security visibility, and automated incident response.

1- First thing to do to perform collection of windows Defender logs

We will turn on the audit file system. To collect Windows Defender's malware detection events, it is necessary to enable the audit file system.

To Do:

Type Local Security Policy → Advanced Audit Policy Configuration → System Audit Policies – Local Group Policy Object → Object Access → Audit File System and change this:



By doing this, we can collect Windows Defender's malware detection events.

NB: Microsoft Defender logs its threat detections and file access events through the Windows Security auditing framework. If file system auditing is off or disabled, Defender does not generate critical events like **file scans**, **quarantines**, or **delete actions**. **For example**, if an attacker accesses sensitive files, our system we look blind during a forensic investigation.

We will configure this auditing baseline first in a standalone lab environment, then scale the deployment for an enterprise architecture.

2- configuration in a small setup.

If we want to enable this detection on one or two machines, how can we do it?

We need to go to Agent configuration (path: `C:\Program Files (x86)\ossec-agent\ossec.conf`) in windows using CLI, run cmd as administrator and type Notepad `C:\Program Files (x86)\ossec-agent\ossec.conf` and then paste this block code in the **log analysis** section

```
<!-- Log analysis -->
<localfile>
  <location>Application</location>
  <log_format>eventchannel</log_format>
</localfile>
```

```
<!-- Start of Windows Defender Log Collection -->
```

```
<!-- Windows Defender Operational Logs -->
```

```

<localfile>
  <location>Microsoft-Windows-Windows Defender/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows Defender WHC Logs -->
<localfile>
  <location>Microsoft-Windows-Windows Defender/WHC</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows VPN Logs -->
<localfile>
  <location>Microsoft-Windows-VPN-Client/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows Security Logs (General threats) -->
<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
  <query>Event/System[Provider[@Name='Microsoft-Windows-Windows Defender']]</query>
</localfile>

<!-- End of Windows Defender Log Collection -->

<localfile>

```

- Explanation of the 4-monitoring channel:

A- <!-- Windows Defender Operational Logs -->

Path: event viewer → Application and services logs → Microsoft → windows → window defender → Operational (we use this name as the monitoring Chanel) → right click on it → select properties → the Full name appears.

```
<location>Microsoft-Windows-Windows Defender/Operational</location>
```

Definition

Windows Defender Operational logs are the primary event logs generated by Microsoft Defender to record its daily runtime activities, security scans, and threat detections.

They track events such as:

- When a malware scan starts, finishes, or is canceled
- Detections of malware, spyware, or potentially unwanted programs (PUPs)
- Actions taken against threats (e.g., cleaning, quarantining, or blocking files)
- Errors or warnings related to the Defender service itself

B- <!-- Windows Defender WHC Logs -->

Path: event viewer → Application and services logs → Microsoft → windows → window defender → WHC (we use this name as the monitoring Channel) → right click on it → select properties → the Full name appears

```
<location>Microsoft-Windows-Windows Defender/WHC</location>
```

Definition

Windows Defender WHC (Windows Health Center) logs are diagnostic records used by Microsoft Defender to report its status and security health to the Windows operating system.

They track operational states such as:

- Whether real-time protection is active
- If virus definitions and security updates are up to date
- The overall health and security posture of the antivirus agent

C- <!-- Windows VPN Logs -->

Path: event viewer → Application and services logs → Microsoft → windows → VPN-Client → Operational (we use this name as the monitoring Channel) → right click on it → select properties → the Full name appears

```
<location>Microsoft-Windows-VPN-Client/Operational</location>
```

Definition

Windows VPN Logs are system event records that track connection activity, status changes, and errors for Virtual Private Network (VPN) connections on Windows.

They track events such as:

- When a VPN connection is initiated, established, or disconnected
- Authentication successes and failures (e.g., incorrect credentials or certificate issues)
- IP address assignments and network routing updates
- Connection drops or errors caused by network interruptions

D- <!-- Windows Security Logs (General threats) -->

Path: Event Viewer (Local) → Windows Logs → Security (we use this name as the monitoring Channel) → right click on it → select properties → the Full name appears

```
<location>Security</location>
```

Definition

Windows Security Logs are centralized audit records managed by the Windows Security auditing subsystem that tracks security-relevant system events, user activities, and threat-related behaviors.

Depending on how your audit policies and endpoint detection agents (like Windows Defender or a SIEM) are configured, they capture events such as:

- Successful and failed user logon attempts and privilege usage
- Account management changes (e.g., user creation or group modifications)
- Object access events (such as file or registry modifications, when auditing is enabled)

- Specific threat indicators when integrated with security tools, such as process creation anomalies, suspicious script executions, or credential-access attempts

Because `Security.evtx` aggregates all general Windows Security logs, the `<query>` tag uses an XPath filter to isolate and forward only events generated by the Microsoft-Windows-Windows Defender provider.

After all these go to PowerShell and run it as administrator and enter the command `Restart-service wazuh` and hit enter

Then

Head over to wazuh manager, go to the agent configuration was made

Threat hunting → event → set it to the last 15min, refresh every 2 seconds

Practice: Now let's perform some activities on the agent: for example, disable our antivirus(turn it off)

Check the wazuh SIEM to see the change

3- Configuration in an enterprise environment.

If we have 100 or 200 machines in an enterprise environment where we want to generate logs or defender logs, we can apply the configuration to all of them through centralized configuration. Keep same structure and make it very simple and professional

Head to wazuh-manager

Click on menu → agent management → select Groups → and groups on which you want to perform modifications;

Here let's do for windows

Windows Groups → files → select agent.conf file → click on Edit agent.conf → add the code block between `<agent_config>` and `</agent_config>` → save

NB: Go to `C:\Program Files (x86)\ossec-agent\ossec.conf` (run cmd as administrator and type `Notepad C:\Program Files (x86)\ossec-agent\ossec.conf`) and remove this, then save, stop wazuh and restart wazuh (`net stop wazuh`, `net start wazuh`)

```
<!-- Start of Windows Defender Log Collection -->

<!-- Windows Defender Operational Logs -->
<localfile>
  <location>Microsoft-Windows-Windows Defender/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows Defender WHC Logs -->
<localfile>
  <location>Microsoft-Windows-Windows Defender/WHC</location>
  <log_format>eventchannel</log_format>
</localfile>
```

```

<!-- Windows VPN Logs -->
<localfile>
  <location>Microsoft-Windows-VPN-Client/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows Security Logs (General threats) -->
<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
  <query>Event/System[Provider[@Name='Microsoft-Windows-Windows Defender']]</query>
</localfile>

<!-- End of Windows Defender Log Collection -->

```

NB: Make sure this agent.conf start with <agent_config> and end with </agent_config> tag

After that, restart wazuh-manager by running all these commands:

```

sudo systemctl restart wazuh-indexer
sudo systemctl restart wazuh-manager
sudo systemctl restart wazuh-dashboard

```

Now let's perform some activities on the agent: for example disable our antivirus(turn it off) or perform scan to see changes

Check the wazuh SIEM to see the change

For Linux add this block code:

```

<agent_config>
<!-- Collect system logs -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/syslog</location>
</localfile>
<!-- Collect authentication logs -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>
<!-- Collect kernel logs -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/kern.log</location>
</localfile>
</agent_config>

```



Microsoft
Defender

Stop Threats. Stay Protected.



Collect Logs



Detect Threats



Monitor Endpoints



Respond Faster

How to Collect Windows Defender and System Logs in Wazuh SIEM

Step-by-Step Guide for Endpoint Visibility, Threat Detection and Incident Response



WAZUH
Open Source Security
for a Safer World

1 Why Collecting Windows Defender Logs Matters?

Collecting Windows Defender logs is important because they provide a critical data source for malware detection, endpoint security visibility, and automated incident response.



Malware Detection
Identify threats early



Endpoint Visibility
See what happens on your systems



Incident Response
Provide forensic evidence



SIEM Integration
Centralized monitoring and alerting

"More visibility. Stronger security."

2 Enable Audit File System (Prerequisite)

To collect Windows Defender's malware detection events, we need to enable the audit file system.

- 1 Open Local Security Policy (Win + R → type sepol.msc → Enter)
- 2 Navigate to: **Advanced Audit Policy Configuration → System Audit Policies → Local Group Policy Object → Object Access → Audit File System**
- 3 Enable Success and Failure
- 4 Click Apply → OK



3 Why It's Important (Details)

- Microsoft Defender logs its threat detections and file access events through the Windows Security auditing framework.
- If file system auditing is off, Defender does not generate critical events like file scans, quarantines, or delete actions.
- For example, if an attacker accesses sensitive files, our system will look blind during a forensic investigation.
- We will configure this auditing baseline first in a standalone lab environment, then scale the deployment for an enterprise architecture.



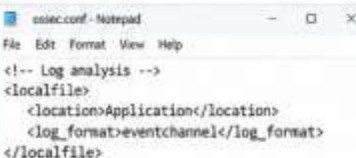
Key Takeaway:

Enable file system auditing to ensure Windows Defender generates the necessary events.

4 Configure in a Small Setup (Single Machine)

- 1 On the Windows agent, open Command Prompt as Administrator.
- 2 Edit the agent configuration file:
`notepad C:\Program Files (x86)\ossec-agent\ossec.conf`
- 3 In the <log analysis> section, add:

```
<!-- Log analysis -->
<localfile>
  <location>Application</location>
  <log_format>eventchannel</log_format>
</localfile>
```



✓ Save the file.

5 Add Windows Defender Log Collection

Add the following XML configuration block to collect Windows Defender logs (Operational, WHC, VPN and Security):

```
<!-- Start of Windows Defender Log Collection -->
<!-- Windows Defender Operational logs -->
<localfile>
  <location>Microsoft-Windows-Windows Defender\Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows Defender WHC logs -->
<localfile>
  <location>Microsoft-Windows-Windows Defender\WHC</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows VPN Logs -->
<localfile>
  <location>Microsoft-Windows-VPN-Client\Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Windows Security Logs (General threats) -->
<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
  <queryevent>System[Process][Name=Microsoft-Windows-Defender][Process]
</localfile>

<!-- End of Windows Defender Log Collection -->
```

✓ Save the file (ossec.conf).

6 Explanation of the 4 Monitoring Channels

6.1 Windows Defender Operational Logs

- Path: Event Viewer → Application and services logs → Microsoft → Windows → Windows Defender → Operational
- Tracks: malware scans, detections, quarantines, cleaning actions, errors, service activity.

6.2 Windows Defender WHC Logs

- Path: Event Viewer → Application and services logs → Microsoft → Windows → Windows Defender → WHC
- Tracks: real-time protection status, updates, and overall health.

6.3 Windows VPN Logs

- Path: Event Viewer → Application and services logs → Microsoft → Windows → VPN-Client → Operational
- Tracks: VPN connection activity, status changes, errors.

6.4 Windows Security Logs (General threats)

- Path: Security (with Microsoft-Windows-Windows Defender provider)
- Tracks: general defense-related security events.

7 Verify Logs in Event Viewer

- 1 Open Event Viewer (Win + R → eventvwr).
- 2 Navigate to: **Application and Services Logs → Microsoft → Windows → Windows Defender → Operational.**



8 Test Detection (Practice)

- 1 After saving the configuration, restart the Wazuh agent service (or Wazuh manager if centralized).
- 2 On the agent, perform an activity, for example: Disable Windows Defender (turn it off).
- 3 Go to the Wazuh dashboard and check for the new logs.



9 Enterprise Environment (Centralized)

For 100 or 200 machines, apply the configuration to all agents using centralized configuration.

- 1 On the Windows agent, open and clean the local ossec.conf (remove local Defender blocks).
- 2 Save the file, then restart the Wazuh agent:
`net stop wazuh`
`net start wazuh`
- 3 Apply the same configuration through Wazuh manager (Agent management → Groups).
- 4 This keeps the structure simple and scalable.



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Stronger Endpoints
Stronger Organizations



Cybersecurity
for a Safer Tomorrow



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Cybersecurity Enthusiast | SOC Content Creator
Learn • Practice • Secure • Share

"Knowledge
Builds a Safer
Tomorrow"